Photometric Test Report

Reference Standards

ANSI C82.77-10-2014
IES LM-79-2019

Prepared For

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Test Laboratory Address:
No.2, Chengwan Road, Suzhou Industrial Park, Suzhou 215122, China
Catalog Number
BPANEL-50S54-UNV-D-8SC61-22-G-WH-H

Project Number
4791498147.3
Report Number
4791498147.3-9a

Test Date
2024-10-30~2024-11-05
Issue Date
2024-11-21
Revision Date
N/A

Prepared By	Approved By
Becky Fan	Roger Xu

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Test List

Sample Received Date: 2024-10-16

Test Item	Sample ID	Model Number	Tests Conducted By
Integrating Sphere Test	7712168	BPANEL-50S54-UNV-D-8SC61-22-G-WH-H	Hom Dong
Goniophotometer Test	7712168	BPANEL-50S54-UNV-D-8SC61-22-G-WH-H	Hom Dong
THD and PF Test	7712168	BPANEL-50S54-UNV-D-8SC61-22-G-WH-H	Hom Dong

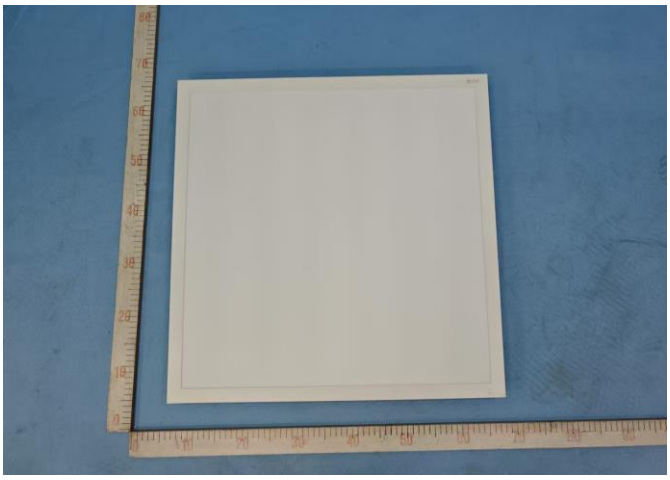
Remark (if any)

1. UL test equipment information is recorded on Meter Use in UL's Aurora database.

Product Description

Luminaire Description: Low Bay Luminaires (Commercial and Industrial)
Model Number: BPANEL-50S54-UNV-D-8SC61-22-G-WH-H
Electrical Ratings and CCT: 120-277Vac, 50/60 Hz, 50W max, 3000K/3500K/4000K/5000K/5700K/6500K tunable
Driver Model Number: BQ-50L6-1250-40-M56
LED Package: BXVN-30E-13H-9NV, BXVN-65E-13H-9NV, Bridgelux Inc
Family Model and Variation: BPANEL-50S54-UNV-D-8SC61-22-XX-YY-H-ZZZ-***-##, where "XX" represent mounting, can be G=Grid mounting, SF=Surface mounting, SP=Suspended mounting, DF=Drywall flange mounting. "YY" represent Finish color, can be WH=White or others. "ZZZ" represent Emergency, can be EM*= Emergency types(* can be blank or any letter), blank=No emergency. "***" represent Sensor receptacle, can be SR#= Sensor receptacle types(# can be blank or any number), blank= Without sensor receptacle. ## represent junction box, can be JB=with junction box, blank=without junction box.

Photos of Products Characteristics



Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-H	Sample ID.	7712168
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

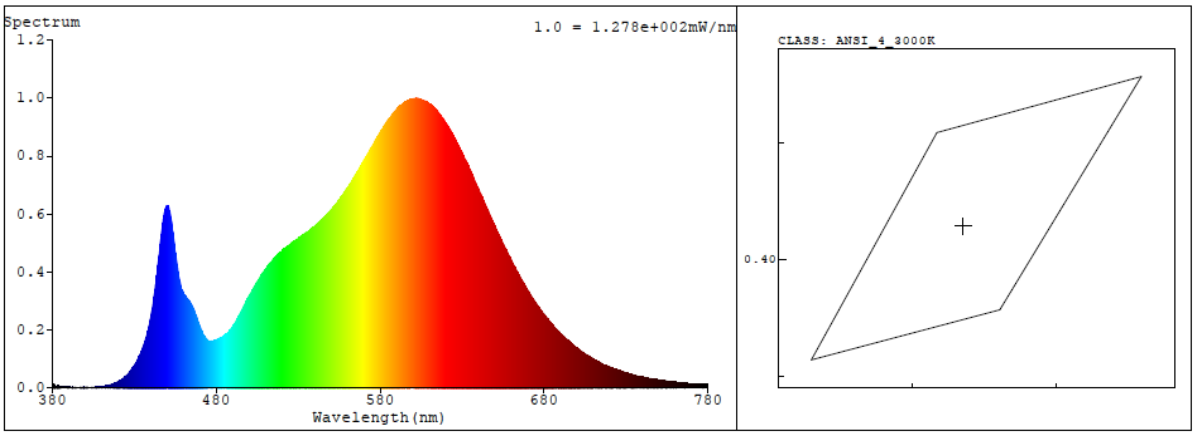
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.405	48.18	0.9903	Horizontal

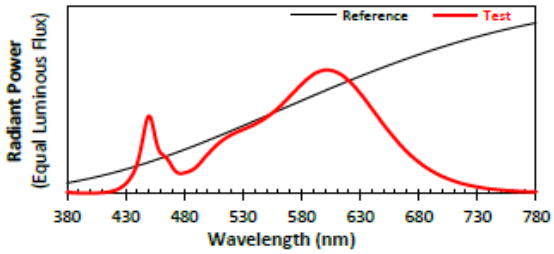
Test Results

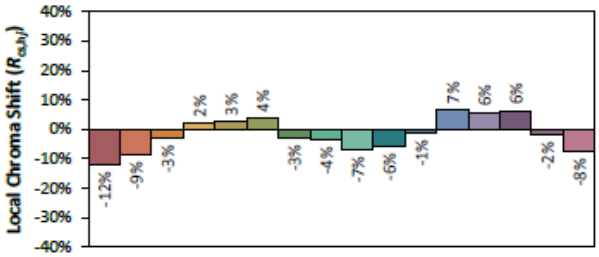
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
3047	82	2	84	96	6292.6	130.6

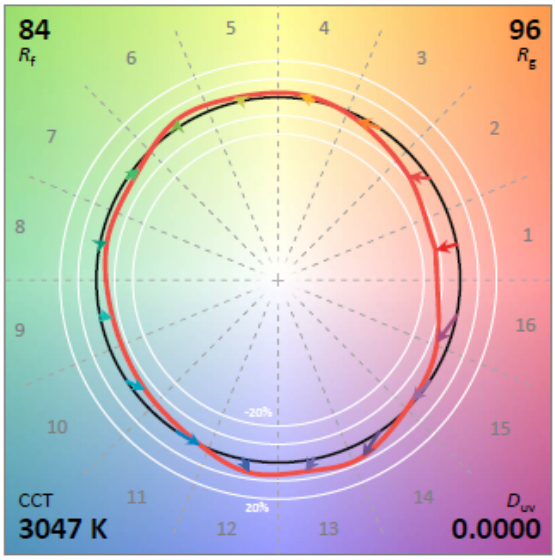
Spectrum

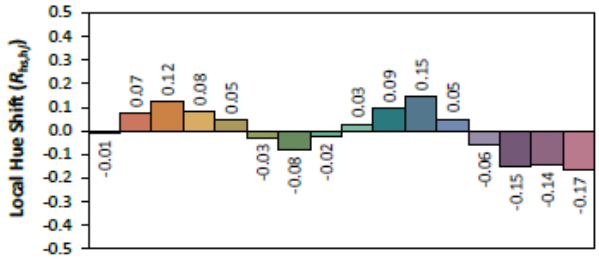


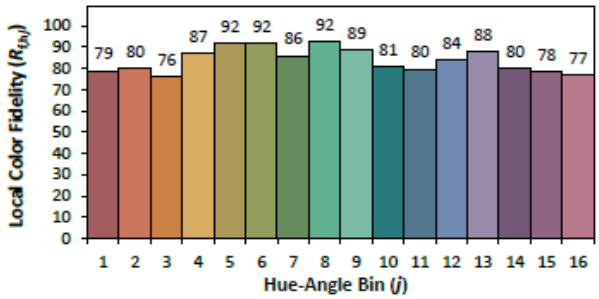
ANSI/IES TM-30-18 Color Rendition Report

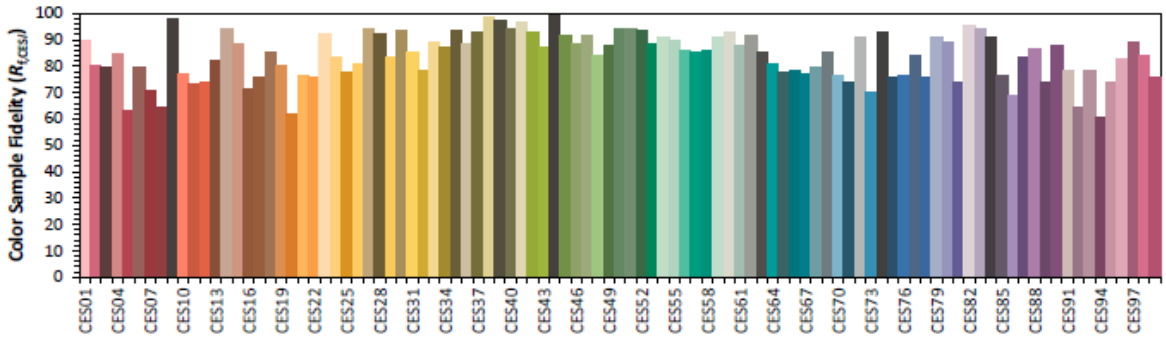












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.4335
y	0.4028
u'	0.2489
v'	0.5204

CIE 13.3-1995 (CRI)	
R_a	82
R_9	2

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-H	Sample ID.	7712168
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

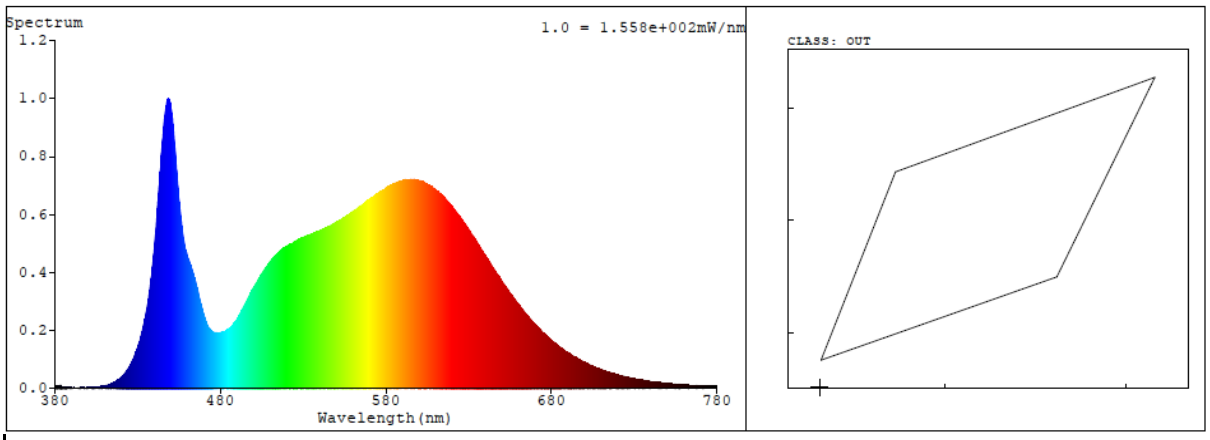
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.404	48.03	0.9903	Horizontal

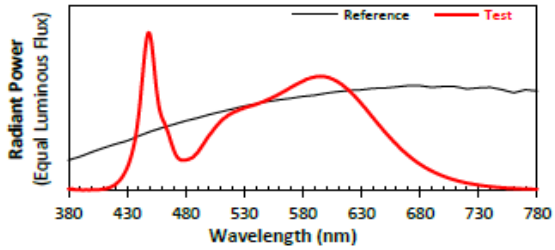
Test Results

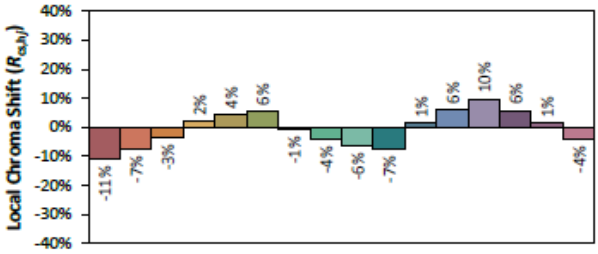
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
4124	84	15	84	98	6489.5	135.1

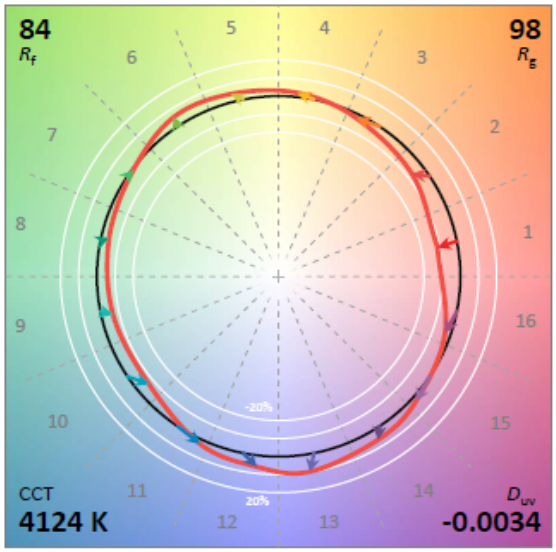
Spectrum

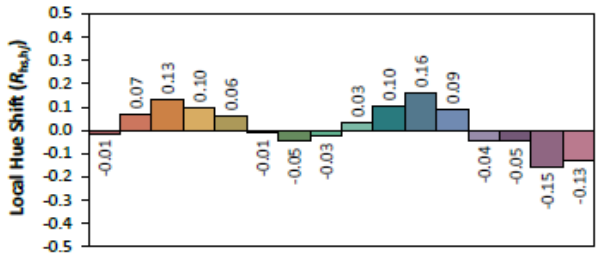


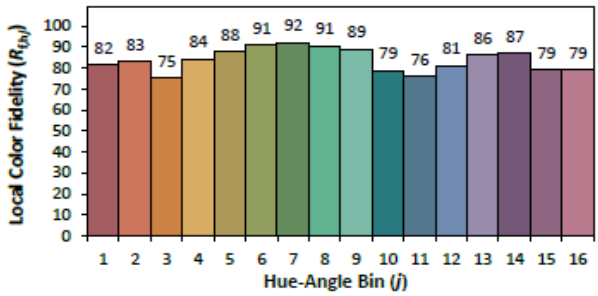
ANSI/IES TM-30-18 Color Rendition Report

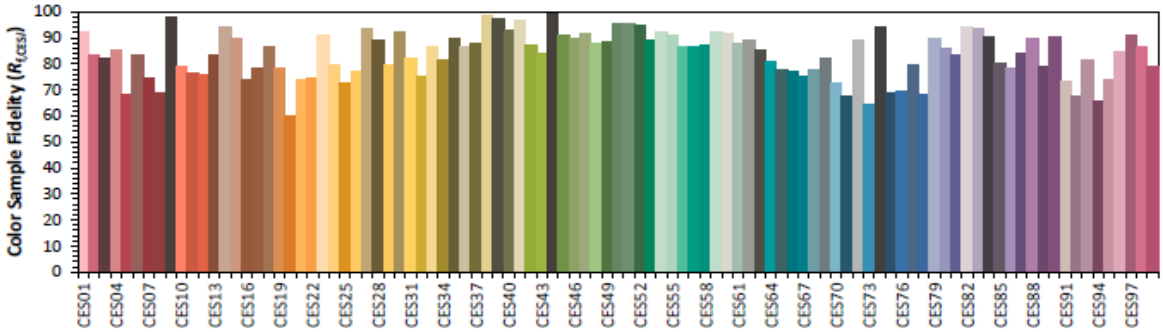












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3731
y	0.3651
u'	0.2249
v'	0.4952

CIE 13.3-1995 (CRI)	
R_a	84
R_9	15

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Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-H	Sample ID.	7712168
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

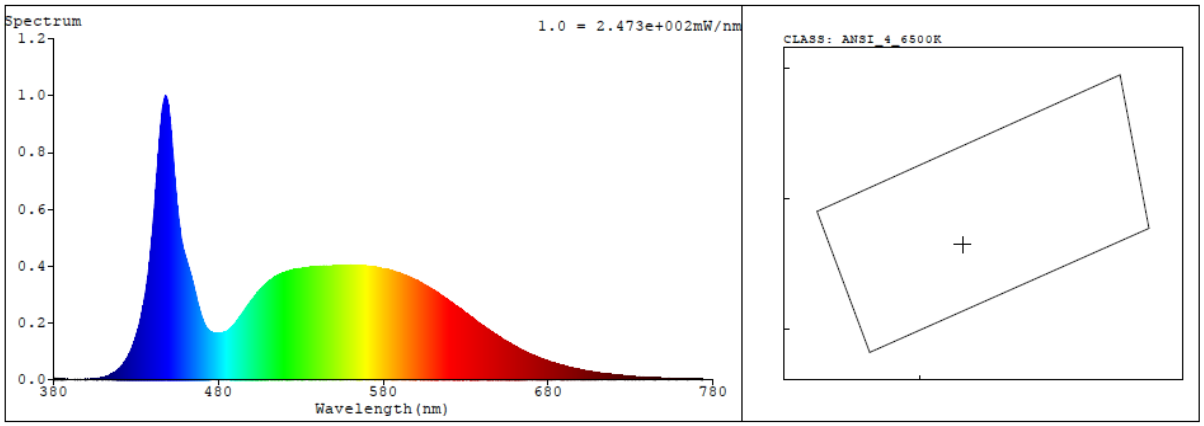
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.406	48.23	0.9905	Horizontal

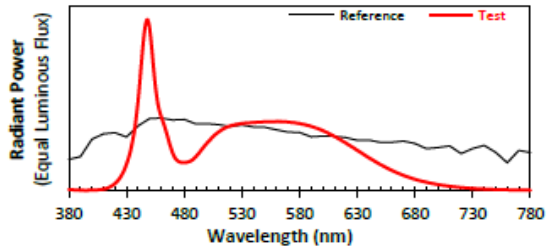
Test Results

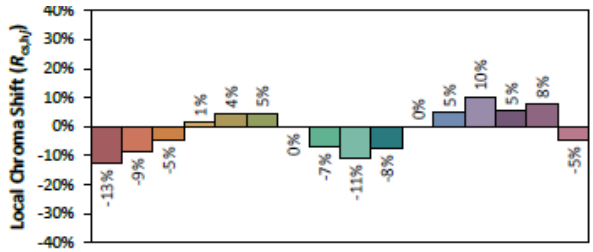
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
6589	81	5	82	96	6484	134.4

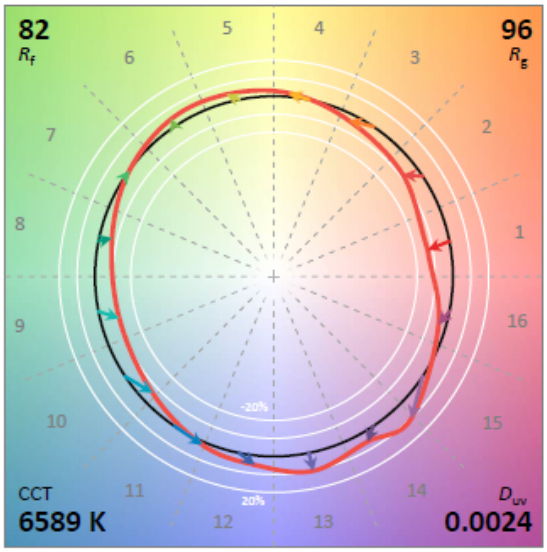
Spectrum

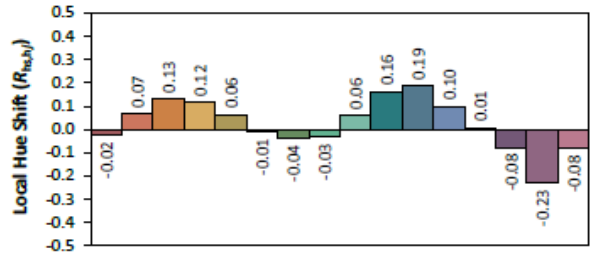


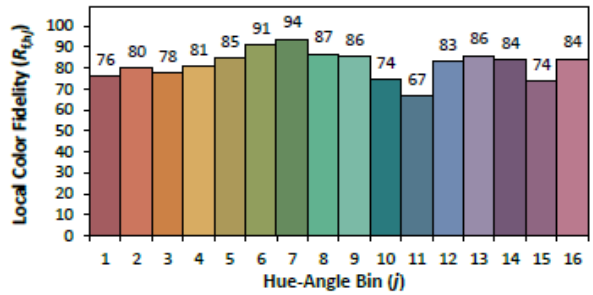
ANSI/IES TM-30-18 Color Rendition Report

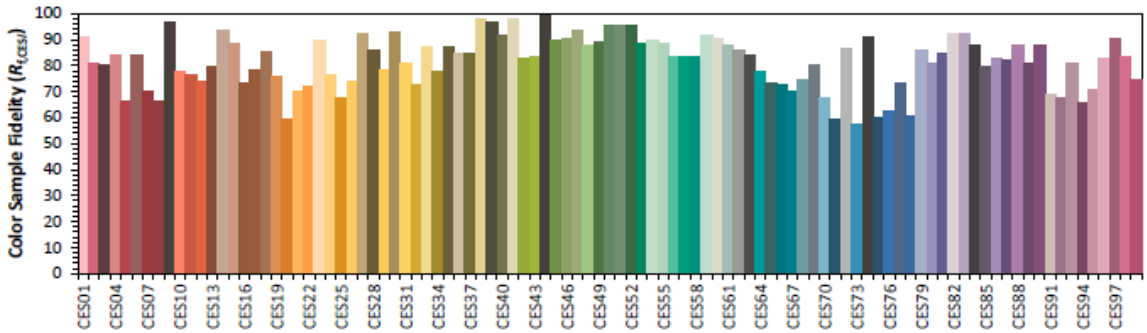












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3116
y	0.3263
u'	0.1981
v'	0.4667

CIE 13.3-1995 (CRI)	
R_a	81
R_9	5

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Goniophotometer Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-H	Sample ID.	7712168
Operate time (Min.)	65	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric parameters were measured using a type C goniophotometer and software.
3.The ambient temperature shall be maintained at 25 °C ± 0.9 °C, measured at a point not more than 1.5 m from the sample and at the same height as the sample. The reference standard lamp is power 400W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
4.The samples were operated at rated voltage and was stabilized before measurement. Luminous flux, luminaire efficacy, zonallumen were calculated from the software taken at 1° vertical intervals and 22.5° horizontal intervals. Photometric distance was more than five times of the largest dimension of the test SSL product.

Goniophotometer Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
24.5	120.09	60	0.406	48.33	0.9922	Horizontal

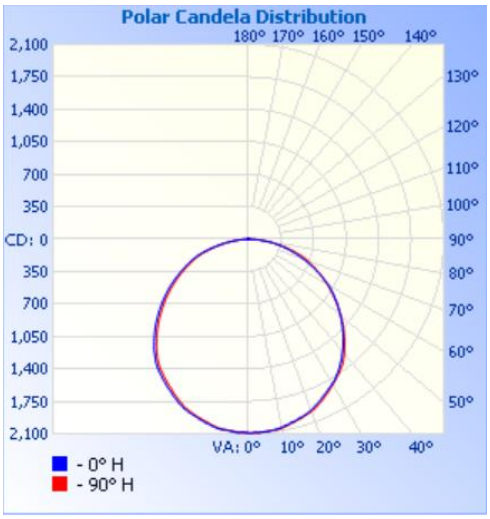
Test Results

Zonal Lumen Requirement (20°-50°)	Field Angle (10%)		Beam Angle (50%)		Flux (lm)	Luminous Efficacy (lm/W)
	Horizontal Spread	Vertical Spread	Horizontal Spread	Vertical Spread		
48.6%	164.8	165	113.6	115.3	6224.1	128.8

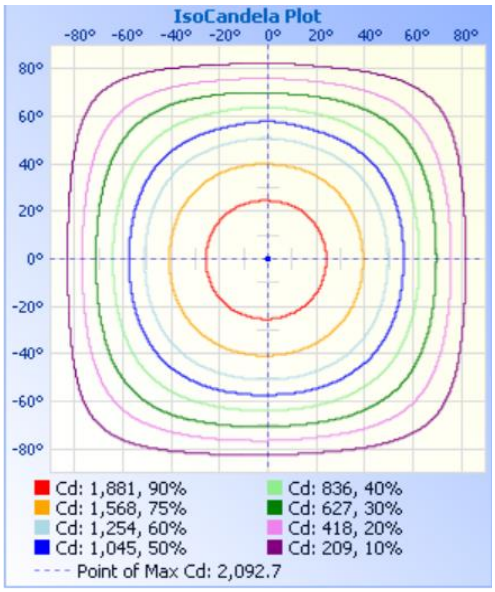
UGR
23.9

Goniophotometer Test (Cont'd)

Polar Candela Distribution



IsoCandela Plot



Goniophotometer Test (Cont'd)

Zonal Lumen Summary

Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	1,634.3	26.3%
0-40	2,694.4	43.3%
0-60	4,785.7	76.9%
60-90	1,422.6	22.9%
70-100	640.6	10.3%
90-120	6.9	0.1%
0-90	6,208.3	99.7%
90-180	15.8	0.3%
0-180	6,224.1	100%

Lumens Per Zone

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	% Total
0-10	198.1	3.2%	90-100	3.1	0%
10-20	568.7	9.1%	100-110	1.9	0%
20-30	867.5	13.9%	110-120	1.9	0%
30-40	1,060.1	17.0%	120-130	2.0	0%
40-50	1,099.0	17.7%	130-140	2.0	0%
50-60	992.3	15.9%	140-150	1.9	0%
60-70	785.0	12.6%	150-160	1.5	0%
70-80	490.9	7.9%	160-170	1.1	0%
80-90	146.6	2.4%	170-180	0.5	0%

Goniophotometer Test (Cont'd)

Intensity Data(cd)

	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5	360
0	2090	2090	2090	2090	2090	2090	2090	2090	2090	2090	2090	2090	2090	2090	2090	2090	2090
1	2092	2088	2086	2091	**2093**	2090	2086	2085	2087	2089	2091	2082	2090	2087	2085	2085	2086
2	2086	2086	2089	2088	2090	2085	2084	2088	2087	2085	2084	2090	2085	2084	2085	2088	2088
3	2090	2088	2085	2089	2089	2087	2087	2088	2091	2089	2084	2085	2085	2084	2082	2086	2085
4	2086	2088	2085	2086	2088	2090	2083	2087	2087	2084	2082	2082	2080	2082	2077	2083	2083
5	2085	2085	2083	2083	2083	2085	2081	2082	2080	2081	2077	2081	2076	2076	2082	2080	2081
6	2083	2078	2079	2080	2081	2085	2078	2077	2077	2078	2075	2074	2071	2072	2074	2079	2076
7	2077	2073	2076	2079	2079	2080	2080	2080	2073	2073	2073	2066	2071	2072	2072	2070	2079
8	2074	2078	2080	2075	2081	2075	2075	2074	2074	2070	2066	2067	2071	2071	2064	2068	2074
9	2067	2067	2072	2077	2076	2076	2075	2066	2069	2063	2060	2065	2061	2061	2064	2069	2067
10	2063	2066	2065	2072	2067	2065	2066	2063	2062	2055	2056	2051	2051	2053	2056	2057	2062
11	2049	2055	2059	2058	2064	2057	2054	2050	2053	2047	2043	2043	2046	2044	2044	2046	2052
12	2045	2043	2046	2049	2052	2051	2047	2045	2041	2035	2036	2035	2030	2031	2036	2041	2039
13	2029	2040	2042	2042	2042	2038	2038	2038	2035	2028	2022	2019	2019	2022	2021	2028	2033
14	2026	2024	2030	2033	2035	2035	2031	2021	2021	2013	2011	2011	2010	2014	2016	2018	2022
15	2016	2020	2024	2022	2022	2021	2017	2017	2015	2005	2001	2000	1999	1998	2000	2006	2016
16	2005	2004	2007	2016	2016	2012	2013	2003	2001	1997	1991	1989	1991	1990	1993	1996	2001
17	1999	2001	2005	2002	2009	2006	2001	1998	1992	1981	1982	1976	1979	1980	1981	1986	1996
18	1988	1990	1990	1997	1993	1993	1996	1988	1982	1978	1969	1966	1969	1968	1974	1977	1983
19	1978	1978	1986	1985	1989	1987	1982	1980	1974	1967	1963	1956	1956	1960	1962	1969	1978
20	1963	1972	1975	1977	1977	1973	1975	1968	1961	1954	1948	1948	1945	1947	1949	1956	1962
25	1880	1894	1895	1899	1905	1901	1894	1887	1882	1872	1861	1859	1857	1858	1863	1874	1885
30	1802	1803	1809	1809	1812	1813	1808	1797	1790	1785	1770	1767	1763	1769	1774	1785	1794
35	1702	1708	1719	1723	1720	1722	1719	1706	1699	1689	1678	1670	1671	1671	1681	1690	1703
40	1574	1592	1601	1601	1605	1597	1588	1587	1576	1559	1545	1540	1538	1542	1552	1567	1581
45	1438	1443	1451	1453	1455	1454	1445	1438	1426	1413	1397	1391	1385	1393	1405	1418	1430
50	1295	1289	1296	1299	1300	1294	1293	1281	1268	1257	1240	1232	1228	1238	1245	1260	1278
55	1147	1129	1139	1140	1139	1137	1133	1120	1105	1096	1080	1071	1065	1084	1085	1100	1111
60	995	971	979	981	981	977	970	959	943	938	917	913	906	923	925	937	950
65	809	811	821	824	823	824	817	801	785	778	761	752	749	770	764	779	791
70	633	667	676	677	676	678	669	651	635	629	615	609	605	614	618	630	645
75	461	476	492	502	504	500	488	476	462	443	438	435	433	428	445	456	469
80	306	294	307	317	318	314	302	289	277	264	258	257	255	259	263	273	284
85	145	138	148	157	158	154	143	130	119	113	105	106	106	115	111	118	126
90	14	18	25	30	32	28	20	10	4	5	5	4	3	8	5	5	7
95	2	2	2	1	2	2	1	2	2	2	3	3	2	2	2	2	2
100	2	1	2	1	1	2	2	2	2	2	2	2	1	2	1	1	1
105	1	2	2	2	2	2	2	2	2	1	1	2	2	2	2	2	2
110	1	2	2	2	2	1	2	2	2	2	1	2	2	2	2	2	2
115	2	2	2	2	2	2	2	2	2	2	3	2	2	2	2	2	2
120	2	2	2	2	1	2	2	2	2	2	2	2	2	3	2	2	2
125	2	2	2	2	2	2	2	2	3	2	2	2	3	2	2	2	2
130	2	2	2	2	3	2	3	3	2	2	2	3	3	2	2	2	3
135	3	3	2	2	2	2	3	3	2	2	3	3	2	2	3	3	3
140	3	3	3	3	3	3	3	3	3	3	3	2	3	3	3	3	2
145	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
150	3	3	3	3	3	3	4	3	3	3	4	4	3	4	3	3	3
155	3	4	3	4	4	4	3	3	4	4	3	3	3	3	3	3	3
160	3	4	4	4	3	3	3	4	3	3	3	3	3	4	4	3	4
165	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
170	4	4	5	5	4	4	5	4	4	4	4	4	4	4	5	4	5
175	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
180	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5

Corrected UGR values generated per CIE 190-2010

UGR Table - Corrected

Reflectances											
Ceiling Cavity		70	70	50	50	30	70	70	50	50	30
Walls		50	30	50	30	30	50	30	50	30	30
Floor Cavity		20	20	20	20	20	20	20	20	20	20
Room Size		UGR Viewed Crosswise					UGR Viewed Endwise				
X=2H	Y=2H	18.6	20.3	19.0	20.6	20.9	18.8	20.4	19.1	20.8	21.1
	3H	20.7	22.2	21.0	22.5	22.9	20.9	22.4	21.3	22.7	23.1
	4H	21.4	22.9	21.8	23.2	23.6	21.7	23.2	22.1	23.5	23.9
	6H	22.0	23.4	22.5	23.7	24.1	22.4	23.7	22.8	24.1	24.5
	8H	22.3	23.5	22.7	23.9	24.3	22.6	23.9	23.0	24.3	24.7
	12H	22.4	23.6	22.8	24.0	24.4	22.8	24.0	23.2	24.4	24.8
4H	2H	19.3	20.7	19.7	21.1	21.5	19.5	20.9	19.9	21.3	21.6
	3H	21.6	22.8	22.0	23.2	23.6	21.8	23.0	22.2	23.4	23.8
	4H	22.5	23.6	22.9	24.0	24.5	22.8	23.9	23.2	24.3	24.8
	6H	23.3	24.2	23.7	24.6	25.1	23.6	24.6	24.1	25.0	25.5
	8H	23.5	24.4	24.0	24.8	25.3	23.9	24.8	24.4	25.2	25.7
	12H	23.7	24.5	24.2	25.0	25.5	24.1	25.0	24.6	25.4	25.9
8H	4H	22.9	23.8	23.3	24.2	24.7	23.2	24.1	23.7	24.5	25.0
	6H	23.7	24.5	24.2	25.0	25.5	24.1	24.9	24.6	25.4	25.8
	8H	24.1	24.7	24.6	25.3	25.7	24.5	25.2	25.0	25.7	26.2
	12H	24.4	25.0	24.9	25.4	26.0	24.8	25.4	25.3	25.9	26.5
12H	4H	22.9	23.7	23.4	24.2	24.7	23.2	24.0	23.7	24.5	25.0
	6H	23.8	24.5	24.3	25.0	25.5	24.2	24.9	24.7	25.3	25.9
	8H	24.2	24.8	24.7	25.3	25.9	24.6	25.2	25.1	25.7	26.3

Maximum UGR = 26.5

THD and PF Test

Model No.	BPANEL-50S54-UNV-D-8SC61-22-G-WH-H	Sample ID.	7712168

Test Method

1. The samples were tested according to the ANSI C82.77-10-2014.
2. The ambient temperature condition was maintained at 25 °C ± 1 °C. The sample measurement was made using a digital power meter and power supply. The sample was operated at rated voltage and stabilized before measurement. The total harmonic distortion were calculated from the digital power meter.

Test Results

CCT Range	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Current THD	Orientation
0%-3000K	120	60	0.406	48.3	0.9922	11.46%	Horizontal
	277	60	0.178	47.3	0.9602	10.85%	Horizontal
20%-3500K	120	60	0.405	48.2	0.9921	10.87%	Horizontal
	277	60	0.177	47.1	0.96	10.18%	Horizontal
40%-4000K	120	60	0.405	48.2	0.9921	10.89%	Horizontal
	277	60	0.177	47.1	0.9600	10.21%	Horizontal
60%-5000K	120	60	0.405	48.2	0.9921	10.87%	Horizontal
	277	60	0.177	47.1	0.9601	10.21%	Horizontal
80%-5700K	120	60	0.406	48.4	0.9921	10.91%	Horizontal
	277	60	0.177	47.2	0.9602	10.24%	Horizontal
100%-6500K	120	60	0.407	48.4	0.9921	10.83%	Horizontal
	277	60	0.178	47.3	0.9605	10.17%	Horizontal

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